How to build a sovereign network? - A proposal to measure network sovereignty

Journal:	*IEEE Communications Magazine*
Manuscript ID	COMMAG-24-00414.R1
Manuscript Type:	Open Call Article
Date Submitted by the Author:	14-Nov-2024
Complete List of Authors:	Janardhanan, Shakthivelu; Technische Universitat Munchen Agarwal, Ritanshi; Universitat der Bundeswehr Munchen Kellerer, Wolfgang; Technische Universitat Munchen Mas-Machuca, Carmen; Universitat der Bundeswehr Munchen
Keywords:	network sovereignty, dependability, measurement metric, manufacturer failure, cut sets
Note: The following files were submitted by the author for peer review, but cannot be converted to PDF. You must view these files (e.g. movies) online.	
ComMag-CSC-ILP.zip	

SCHOLARONE™
Manuscripts

How to build a sovereign network? - A proposal to measure network sovereignty

Shakthivelu Janardhanan, *Student Member, IEEE,* Ritanshi Agarwal, Wolfgang Kellerer, *Senior Member, IEEE,* Carmen Mas-Machuca, *Senior Member, IEEE.*

***Abstract*—Network sovereignty is a network operator's ability to reduce the dependency on component manufacturers to minimize the impact of manufacturer failures. Network operators now face new design challenges to increase network sovereignty and avoid vendor lock-in problems because a high dependency on a manufacturer corresponds to low survivability if that manufacturer is unavailable. The main contribution of this work is the proposal of a novel metric to measure network sovereignty, the Cut Set Coloring (CSC) score. Based on the CSC core metric *CSC-ILP*, our Integer Linear Program formulation is presented to maximize network sovereignty. We compare *CSC-ILP*'s performance with state of the art manufacturer assignment strategies.**

***Index Terms*—network sovereignty, dependability, measurement metric, manufacturer failure, cut sets**

I. Introduction

Though network operators can buy components from different manufacturers, they favor only one or two manufacturers to ease interoperability while saving costs on bulk purchasing and deployment. However, this strategy causes a high dependency on them. If one of those manufacturers becomes unavailable due to inherent component design flaws, geopolitical ban, subcomponent manufacturers becoming unavailable, security vulnerabilities, etc., all components of that manufacturer are compromised simultaneously. We define a *manufacturer failure* as the simultaneous unavailability of all components from that manufacturer due to any of the aforementioned reasons. Operators aim to reduce the impact of such massive correlated simultaneous failures.

Several examples of different types of such manufacturer failures have been recorded. It can be a natural disaster or an accident like the fire that burned down the OVHcloud data centers in 2021. An estimated 3.6 million websites and 464,000 domain names were affected [1], causing an estimated loss of €105 million. From 2014 to 2023, [1] has identified 22 major data center fires. Such accidents furthered the cause for backup websites and data storage hosted on different cloud platforms to avoid a singular dependency. Such failures can also be inadvertently man-made. The recent CrowdStrike outage [2] of 2024 caused an estimated 8.5 million devices to fail simultaneously, resulting in losses of over $5.4 billion in revenue. This was caused by a singular dependency on CrowdStrike and Windows. Such failures can also be security vulnerabilities, as recorded in the CVE vulnerabilities list [3]. For instance, CVE-2021-0275, a cross-site scripting vulnerability, affected Juniper Networks Junos Operating System (OS) [3], which is a popular OS for enterprise and Data Center Networks (DCNs). Similarly, CVE-2013-6026 was a firmware backdoor that allowed attackers to remotely access several D-Link, Alpha Networks, and Planex routers [3]. Such security issues form a shared vulnerability because all the routers from a vendor use the same firmware.

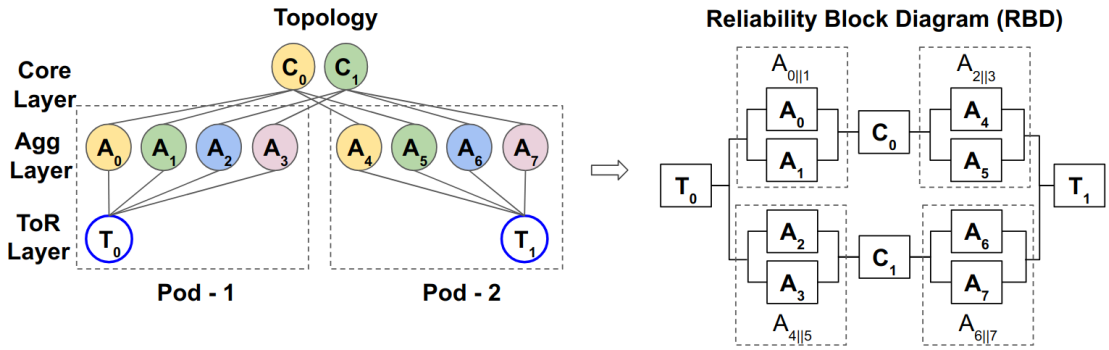
Recently, governments are removing foreign dependencies from core infrastructure. For example, the United States, Spain, England, Germany, etc. have banned Chinese equipment from their core networks [4]–[6]. [7] details the policy-making trends on digital autonomy in several countries, like Russia's sovereign network (RuNet) and financial services via Mir [4] to the European Union's GAIA-X initiative, to reduce the dependency on American cloud services. These policies aim to remove foreign dependencies and vendor-related threats in the critical infrastructure.

A system's availability is the probability that it delivers uninterrupted service at any instant [8]. Extending this definition, a network's availability is the probability that it can route all its flows at any instant. In this work, a flow refers to a unit of traffic between a source and destination. A networking device's availability is provided based on stress tests on the device in its datasheet through parameters such as mean time to fail (MTTF), mean time to repair (MTTR), etc. The availabilities of all the devices in the network can be used to derive a network's availability using tools like the Reliability Block Diagram (RBD).

Network availability is often confused with sovereignty. To understand the difference, let us consider the following example with a simplified DCN topology with twelve nodes, as shown in Fig. 1.

Let us first compare the availability and sovereignty for the DCN when all twelve nodes are from a Manufacturer-A (M_A) with node availabilities (A_n) of 0.999. Let there be a flow between nodes T_0 and T_1. This flow's availability (A_f) can be evaluated using the RBD in Fig. 1. When elements in an RBD are in series, their availabilities are multiplied, while parallel elements are combined as the probability that at least one of them is working. In Fig. 1, (i) A_0, A_1, A_2, and A_3 are in parallel, (ii) C_0 and C_1 are in parallel, (iii) A_4, A_5, A_6, and A_7 are in parallel, and (iv) the subsystems in (i), (ii), and (iii) are in series with the source (T_0) and destination (T_1) nodes. Using the equation for A_f in Fig. 1, A_f is calculated as 0.998.

This work has received funding from the Bavarian Ministry of Economic Affairs, Regional Development, and Energy under the project "6G Future Lab Bavaria" and by the Federal Ministry of Education and Research of Germany in the program of "Souverän. Digital. Vernetzt.". Joint project 6G-life- 16KISK002.



$$A_{x||y} = 1 - (1 - A_{A_x})(1 - A_{A_y})$$

$$A_f = A_{T_0} \times \left[1 - \left(1 - \left(A_{0||1} \cdot A_{C_0} \cdot A_{4||5}\right)\right)\left(1 - \left(A_{2||3} \cdot A_{C_1} \cdot A_{6||7}\right)\right)\right] \times A_{T_1}$$

Fig. 1: Network sovereignty vs. availability example, using a DCN topology and its RBD. A_f and A_n denote the flow and node availabilities, respectively. $A_{x||y}$ denotes the availability of two nodes x and y in parallel. Each node color represents a different manufacturer.

However, if M_A fails, all the nodes in the network fail, and the flow also fails due to the overwhelming dependency on M_A. Hence, this case has minimal sovereignty.

Let us consider the second case, where the nodes are from four different manufacturers- M_A (yellow), M_B (green), M_C (blue), and M_D (pink) with A_n of 0.999, 0.995, 0.990, and 0.985, respectively. Let T_0 and T_1 be from M_C and M_D, respectively. Now, with the same RBD but different A_n, A_f drops to 0.975. However, if one manufacturer (for example, M_A) fails, several working paths remain between T_0 and T_1. Therefore, the dependency on M_A is avoided by introducing more manufacturers. This case shows higher sovereignty but lower availability than the first example.

The above example shows that dependability attributes like availability cannot represent the impact of manufacturer failures. To address this shortcoming, technology sovereignty [9] was defined as an organization's ability to provide a technology without any dependencies on suppliers. Extending to communication networks, we define network sovereignty as a network operator's ability to operate a network without any dependencies on component manufacturers.

Given a network topology, an operator must decide how many manufacturers to buy nodes from and where to place these nodes. The research question in this work is the Manufacturer Assignment for Sovereignty (MAS) problem [5], [10]:

(Q) Given a network topology as a set of interconnected nodes and the number of node manufacturers, what is the best manufacturer assignment possible to maximize network sovereignty? In other words, how many nodes should be purchased from each manufacturer, and where must they be placed in the topology?

The contributions of this work to that question are as follows.

1) Introducing a novel network sovereignty metric- 'Cut Set Coloring (CSC) score' (Section III).
2) Evaluating the most sovereign manufacturer assignment with our CSC score-based ILP formulation, *CSC-ILP* to solve the MAS problem (Section III-D).
3) Comparing *CSC-ILP*'s performance with centrality metrics-based heuristics and state of the art works like *Diversity Assignment Problem (DAP)* [11], *Min cut set heuristic* [10] and *Naga* [5] (Section IV).
4) Deriving design guidelines for network sovereignty (Section V).

II. Sovereignty and Multi-Vendor Networks in Literature

Recent trade wars, sanctions, and economic crises highlight today's need for technology, data, and network sovereignty.Network sovereignty differs from network redundancy and diversity [4], [5]. Redundancy by duplicating network elements to increase network paths may improve availability. However, sovereignty is not improved if all components are from the same manufacturer. In contrast, network diversity provides multiple technology and/or operator-independent paths to connect to the network. However, if the same operator provides both technologies and/or the operators use components from the same manufacturer, sovereignty is not improved. Therefore, network sovereignty aims to provide multiple technology, operator, and manufacturer-independent paths in the network.

Common issues in establishing sovereignty are geopolitics, multi-layer-multi-domain security, and lack of quantification metrics [4], [5], [10]–[12]. [11] deals with the *Diversity Assignment Problem (DAP)* for improved security and resilience. *DAP*-ILP aims to introduce diversity among homogeneous routing nodes by introducing 'variant nodes' to avoid shared vulnerabilities between the nodes. Since DAP aims to remove dependencies on particular node configurations, [11] can be viewed as a work on network sovereignty and compared against our *CSC-ILP*. Similarly, [12] also employs node heterogeneity through the *(r, s)-robustness with coloring* method to improve resilience against adversaries. However, [11], [12] do not scale well with the network size.

Our previous work [10] on network sovereignty proposes a heuristic manufacturer assignment, called the *Min cut set*

heuristic, based on minimal cut sets (MCSs). A cut set of a flow in a network is a set of elements that, when failed, causes the flow to fail. A cut set is minimal if it can not be reduced further while still being a cut set. For example, in the topology in Fig. 1, (A_0, A_1, A_2, A_3) is a MCS. The source and destination are discarded as MCSs because if they fail, the flow fails [5], [10], [11]. The *Min cut set heuristic* solves the MAS problem by maximizing the number of manufacturers (manufacturer diversity) inside the MCSs for each flow in the network to avoid a single point of failure. However, [10]–[12] can not quantify a network's sovereignty.

We proposed the Path Set Diversity (PSD) score metric [5] to measure and compare the network sovereignty of different manufacturer assignments. *Naga*, an ILP formulation, was also proposed to solve the MAS problem. The PSD score for each path of a flow is equal to the reciprocal of the number of the number of manufacturers in that path. This is because, several manufacturers in a path denotes a dependence on all of those manufacturers for the flow to be successful. For example, in Fig. 1 with four manufacturers, the path-PSD score for the path T_0-A_0-C_0-A_4-T_1 is 1 because only yellow nodes are present in the path. The path-PSD score for the path T_0-A_1-C_0-A_4-T_1 is $\frac{1}{2}$ because there are two manufacturers in the path (green and yellow). The flow-PSD score is equal to the sum of all the path-PSD scores. However, paths with the same manufacturer combination should be counted only once. For example, the paths T_0-A_2-C_1-A_7-T_1 and T_0-A_3-C_1-A_6-T_1 have the same manufacturers in the path (blue, green, and pink). The order and the number of nodes in the path are irrelevant for a sovereignty study. The network-PSD score is the average of all the flow-PSD scores. *Naga* [5] maximizes the network-PSD score to solve the MAS problem.

In Fig. 1, there are eight simple paths between T_0 and T_1. However, the number of simple paths for each flow in a network can be tens of thousands. Additionally, there are several flows in a network adding to the problem complexity. Therefore, *Naga* considers only k shortest paths, where choosing an appropriate k is a challenge. *Naga* does not scale for large networks and/or a large k. Furthermore, choosing the k shortest paths for a symmetric topology with comparable link lengths, like the DCN topology in Fig. 1, is ambiguous. If the paths are not prioritized based on link lengths, the PSD score may not quantify the network's sovereignty accurately and *Naga* may not produce the optimal manufacturer assignment.

In this work, we propose the novel CSC score metric to measure and compare the network sovereignty of manufacturer assignments while ensuring scalability and application on asymmetric and symmetric topologies.

III. Cut Set Coloring (CSC) score as a network sovereignty metric

A. Towards a sovereign network

A minimal cut set (MCS) gives the set of components that, when failing, causes the flow to fail. If all the MCS's nodes are from the same manufacturer, and if that manufacturer fails, the MCS fails, causing the flow to fail. Therefore, maximizing manufacturer diversity within an MCS is necessary to avoid a dependency on one manufacturer. Hence, the main guideline is: *Each MCS must have nodes from as many different manufacturers as possible to prevent the simultaneous failure of all the nodes in the MCS.*

For example, consider the MCS (A_0, A_1, A_2, A_3) in the topology in Fig. 1. If the MCS's nodes are from the same manufacturer (first case in Section I), the MCS fails if that manufacturer is unavailable, causing flow failure. However, if the MCS's nodes are from different manufacturers (second case in Section I), then the MCS fails only if all the manufacturers in the MCS fail simultaneously. However, the simultaneous failure of multiple manufacturers is unlikely. Hence, such a diversified manufacturer assignment improves sovereignty by removing the dependency on manufacturers. The aforementioned guideline must be followed for every MCS of each flow in the network.

B. Cut Set Coloring (CSC) score

The MCS's CSC score, called the MCS-CSC score, is qualitatively defined as the extent of manufacturer diversity in that MCS. If there is only one manufacturer, the MCS-CSC score should be penalized with a score of 0. If there is more than one manufacturer in the MCS, the singular dependency is avoided, and this is rewarded by a score equal to the number of manufacturers in the ith MCS ($|M^i|$) divided by the number of nodes in the MCS ($|n^i|$). If all the MCS's nodes are from different manufacturers, then the MCS has the highest MCS-CSC score of 1. Quantitatively,

$$\text{MCS-CSC score} = \begin{cases} 0, & \text{if} |M^i| = 1 \\ \frac{|M^i|}{|n^i|}, & \text{otherwise} \end{cases}. \tag{1}$$

A flow is only as sovereign as its weakest MCS [4]. Therefore, the flow's CSC score called the Flow-CSC score, is equal to the least MCS-CSC scores of all the MCSs of that flow. The Network-CSC score combines all the Flow-CSC scores in the network, represented as a range in a box plot. It can not be denoted by a single value, for example, as an average of all the Flow-CSC scores, because information on the weakest Flow-CSC score and the variance of the Flow-CSC scores might get lost. For example, two assignments may have the same mean values, but one assignment may have a worse weakest Flow-CSC score and must be avoided.

C. Application of the CSC score

Consider the example DCN topology in Fig. 1. Consider four manufacturers available to the operator. Consider a flow between T_0 and T_1. The MCSs for this flow are $\{(A_0, A_1, A_2, A_3), (A_0, A_1, C_1), (A_0, A_1, A_6, A_7), (C_0, C_1), (C_0, A_2, A_3), (C_0, A_6, A_7), (A_4, A_5, C_1), (A_4, A_5, A_6, A_7)\}$. For the MCSs (A_0, A_1, C_1) and (A_4, A_5, C_1), only two manufacturers (yellow and green) are present for three nodes. Hence, from Eq. 1, they have MCS-CSC scores of $\frac{2}{3} = 0.66$ each. For each of the other MCSs, the number of manufacturers ($|M^i|$) equals the number of nodes ($|n^i|$). Therefore, they all have a score of 1 each. For this flow, the minimum MCS-CSC score is 0.66. Therefore, the Flow-CSC

score is also 0.66. For a network, the Network-CSC score is represented as a range of Flow-CSC scores calculated for each flow.

D. Problem formulation: CSC-ILP

Consider a network topology, with vertices V and edges E. Let each flow be defined by its source and destination. We remove all the flows with a one-hop path because they do not have any other MCSs. Let the available manufacturer list be M. Each flow has its set of MCSs. The order of the MCSs is irrelevant.

CSC-ILP solves the MAS problem in (Q) by assigning each node to one manufacturer such that the network sovereignty is maximized. For this purpose, *CSC-ILP* identifies the minimum MCS-CSC score among all the flows in the network. Then, this MCS-CSC score is maximized by increasing the number of manufacturers in that MCS. This global optimization ensures the improvement in all the flows and, thereby, the network's sovereignty.

As a secondary objective to improve the results, the sum of all the flows' MCS-CSC scores is maximized. This maximizes the diversity in as many MCSs as possible.

For a network operator, the most important parameter is cost. Each manufacturer's node has a different cost. Therefore, *CSC-ILP* considers the operator's cost constraint. The cost assumptions in this work are discussed in Section IV-A.

IV. Case studies

A. Experimental setup

We evaluate the CSC score metric and *CSC-ILP* on four core networks [13]: (i) Abilene ($|V| = 11$ (one node with nodal degree of one is removed), $|E| = 14$), (ii) Polska ($|V| = 12$, $|E| = 18$), (iii) Germany_17 ($|V| = 17$, $|E| = 26$), and (iv) Nobel_EU ($|V| = 28$, $|E| = 41$). We consider any-to-any traffic and four different numbers of manufacturers $(2, 3, 4, 5)$ in the market to compare varying degrees of competitive markets.

To include the operator's cost constraint as per Section III-D, in this work, we assume that the different manufacturers' node costs (C_m) vary linearly between 0.995 and 0.985. For example, if there are two manufacturers, then C_0 and C_1 are 0.995 and 0.985, respectively. If there are three manufacturers, then C_0, C_1, and C_2 are 0.995, 0.99, and 0.985, respectively. Since there is no operator cost budget data available, we assume that the cost threshold (C_T) is equal to the product of the number of nodes in the network ($|V|$) and the mean possible cost per node, i.e., $|V| \times \frac{0.995+0.985}{2}$. C_m and C_T are *CSC-ILP*'s input parameters that the operator must update as per their data.

Though we evaluate core networks, the CSC score metric and *CSC-ILP* apply to any network.

B. Physical setup and run time

CSC-ILP uses Python-Gurobi optimization suite [14] and was run on an AMD Ryzen 3700X octa-core processor with 32GB RAM. *CSC-ILP*'s run time for Nobel_EU for five manufacturers was around 2 minutes, showing strong scope for scalability. On the same physical setup, *DAP* took five hours for Nobel_EU for four manufacturers, and did not deliver a result for five manufacturers after eight hours, highlighting the scalability issue discussed in [11].

C. State of the art to compare with CSC-ILP

Different solutions have been evaluated and compared with *CSC-ILP*.

Centrality metrics evaluate a node's connectivity characteristics. Higher centrality means the node is more important. If an operator assigns all the important nodes to the same manufacturer and it fails, network operation is greatly affected. Therefore, the operator would assign nodes of equal importance to different manufacturers. The same logic applies to the centrality metrics-based assignments. The nodes are arranged in a sorted descending list according to their centrality metric values. Then, the nodes are assigned alternatively to different manufacturers. For example, consider the topology in Fig. 1 and the nodal degree (ND) centrality metric. The nodes' NDs are $\{\text{ND}(A_0) = 2,\ \text{ND}(A_1) = 2,\ \text{ND}(A_2) = 2,\ \text{ND}(A_3) = 2,\ \text{ND}(A_4) = 2,\ \text{ND}(A_5) = 2,\ \text{ND}(A_6) = 2,\ \text{ND}(A_7) = 2,\ \text{ND}(C_0) = 4,\ \text{ND}(C_1) = 4\}$. The source and destination are disregarded in this sovereignty analysis. The resulting sorted list in descending order of ND is $\{C_0, C_1, A_0, A_1, A_2, A_3, A_4, A_5, A_6, A_7\}$ (nodes with same ND can be written in any order). If there are two manufacturers, the nodes $\{C_0, A_0, A_2, A_4, A_6\}$ and $\{C_1, A_1, A_3, A_5, A_7\}$ are from the first and second manufacturers respectively. Similarly, if there are three manufacturers, then $\{C_0, A_1, A_4, A_7\}$, $\{C_1, A_2, A_5\}$, and $\{A_0, A_3, A_6\}$ are from the first, second, and third manufacturers respectively. Hence, the nodes of similar ND are distributed among different manufacturers. A comprehensive comparison is evaluated considering three different centrality metrics- Nodal Degree (ND), Betweenness Centrality (BwC), and Closeness Centrality (CC) [5], [10].

Furthermore, manufacturer assignments from *DAP* [11], *Naga* [5], and *Min cut set heuristic* [10] are also compared against the *CSC-ILP* to understand the improvement with *CSC-ILP*.

D. Evaluation Procedure

The assignments for different topologies, numbers of manufacturers, and strategies like *CSC-ILP*, *DAP*, *Naga*, *Min cut set heuristic*, ND, BwC, and CC are obtained. First, the Network-CSC scores of all these assignments are calculated.

Second, we perform all possible manufacturer failure combinations sequentially. For example, when there are three manufacturers, the possible failure combinations of the manufacturers 0, 1, and 2, are $\{(0), (1), (2)(0, 1), (1, 2), (0, 2)\}$, where, $(0, 1)$ is when manufacturers 0 and 1 have failed simultaneously. It is not meaningful to consider $(0, 1, 2)$, the case where all manufacturers fail simultaneously. Then, we evaluate the Average Two Terminal Availability (ATTA). ATTA is the probability that a particular source-destination pair is connected [5], [11], [15]. We represent the ATTA as the percentage of successful flows under different failure

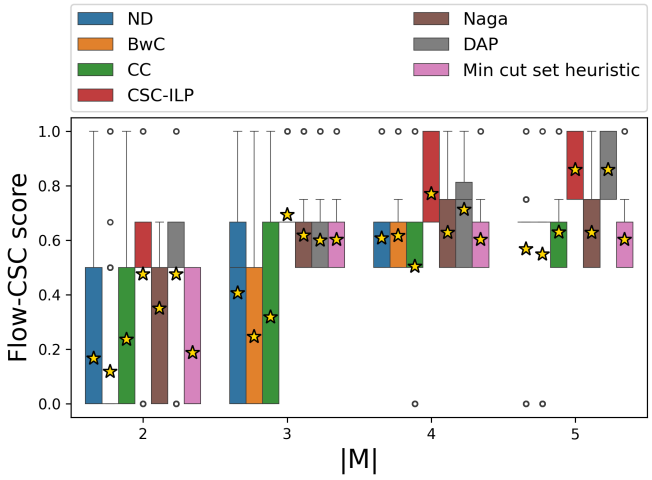


Fig. 2: Network-CSC score represented as Flow-CSC score distribution vs. Number of manufacturers ($|M|$) for Polska topology. The proposed *CSC-ILP* assignments outperform *DAP*, *Naga*, *Min cut set heuristic*, and centrality metrics. Flow-CSC score increases with more manufacturers. The star in the box plots represents the mean. The heights of different color bars indicate the distribution of the central 50% of the Flow-CSC scores. The lines on the top and the bottom of the boxes denote the remaining data range on the top and bottom quartiles, respectively. The white circles denote outliers. The absence of a box shows that the Flow-CSC scores are concentrated near the mean.

scenarios. When a network is able to route all the flows, if at least any one manufacturer is operational, it can be called a fully sovereign network. However, this is unlikely in a real-world network.

V. Case studies- Results and Guidelines

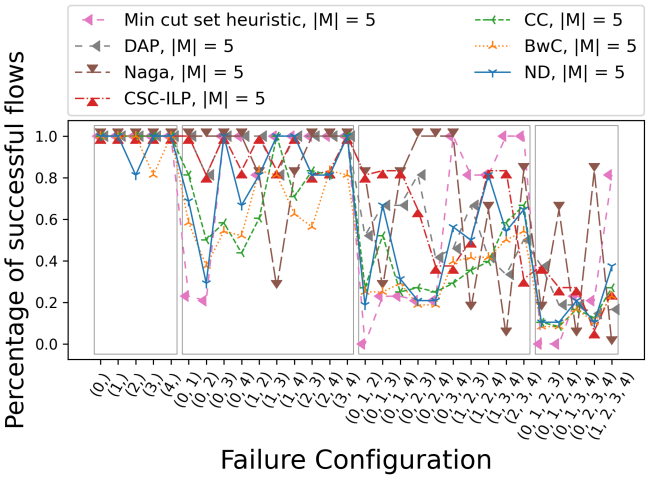
A. Evaluation of the CSC score and CSC-ILP

In this section, we convey that

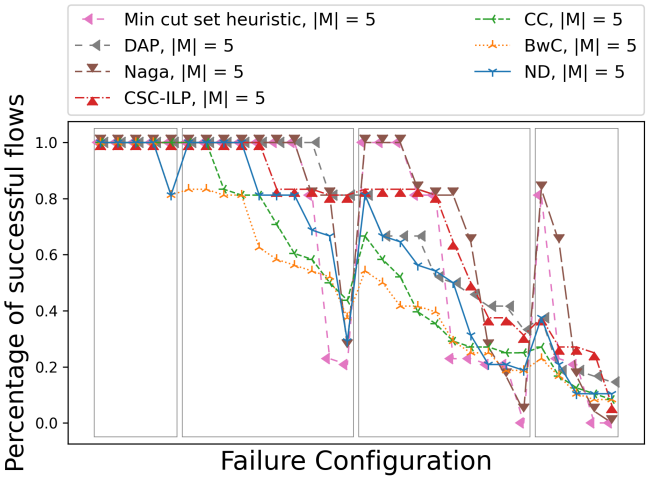
(V1) our proposed Network-CSC score is a strong indication of a network's sovereignty and

(V2) *CSC-ILP* outperforms the other assignments in comparison.

To show that the Network-CSC score is a strong indication of a network's sovereignty, we show that the successful flows under failure scenarios are the highest for *CSC-ILP*. Fig. 2 shows the Network-CSC score as a box plot of all the Flow-CSC scores in the network along the Y-axis, for the manufacturer assignments from *CSC-ILP*, *Naga*, *Min cut set heuristic*, *DAP*, ND, BwC, and CC for the Polska topology. In some cases, the star denoting the mean Flow-CSC score is outside the box because the mean value is skewed by an outlier. This is why the Network-CSC score should not be represented as a single value since the distribution data can be lost. The X-axis shows the different possible numbers of manufacturers ($|M|$) that could be available in the market for the operator.



(a) Percentage of successful flows (ATTA) vs. Failure scenarios



(b) Percentage of successful flows (ATTA) in descending order, grouped by the number of failing manufacturers

Fig. 3: *CSC-ILP*'s performance: Polska with five manufacturers ($|M| = 5$), *CSC-ILP* outperforms other assignments.

There are two important features to note in Fig. 2- (i) the assignment with the best average, and (ii) the assignment with the best worst-case performance, i.e., the assignment which has the highest Flow-CSC score even in the worst case. Irrespective of $|M|$, the *CSC-ILP* performs best in both the aforementioned features. The average Flow-CSC score (star) and the lowest Flow-CSC score for the *CSC-ILP* for each $|M|$ is always equal or better than that of the other assignments. Therefore, *CSC-ILP* outperforms the other manufacturer assignments and maintains the best possible worst-case Flow-CSC scores. *DAP* performs close to *CSC-ILP*. This is because *DAP* provides the most optimal assignment to improve the average ATTA.

Fig. 3a represents ATTA by plotting the percentage of successful flows on the Y-axis against the failure combinations from Section IV-D, along the X-axis, for the Polska topology for five manufacturers. Each line corresponds to a particular

assignment. To easily compare the assignments' ATTA, we plot Fig. 3b based on Fig. 3a. We group the failures into the grey boxes according to the number of failing manufacturers. Then, inside each group, we rearrange the lines in descending order.

From Fig. 3, ATTA decreases when more manufacturers fail. Moreover, *CSC-ILP* outperforms the centrality metrics-based heuristics. At first glance into Fig. 3b, *DAP*, *Min cut set heuristic*, and *Naga* seem to outperform *CSC-ILP* in some cases. However, *CSC-ILP* maintains the best possible worst-case ATTA. For example, for the two-manufacturer failure scenario (grey box in the middle) in Fig. 3b, the worst possible ATTA for *CSC-ILP* and *Naga* is 0.45 and 0.25 respectively. Therefore, *CSC-ILP* minimizes the impact of the worst-case manufacturer failures as expected. *DAP* provides strong ATTA results because it is globally optimized to maximize average ATTA in the network. However, it has weaker worst-case performance and is computationally very intensive. *DAP*'s scalability issue is addressed in [11] and the authors propose heuristics to be feasible.

The results observed are consistent for all the sample topologies. Therefore, the graphs shown in Figs. 2 and 3 support the statements (V1) and (V2). Now, we can proceed with our Network-CSC score analysis.

B. Analysis and Inferences from the Network-CSC scores

This section answers important questions from an operator's standpoint.

1) What is the impact of purchasing components from more manufacturers?
Fig. 2 shows that more manufacturers increase the Flow-CSC scores because when more manufacturers are available, the dependency on one manufacturer is reduced. Therefore, the network sovereignty increases. Using *CSC-ILP* with more manufacturers is more beneficial to the operator.

2) How many manufacturers does my network require?
The ideal number of manufacturers required for a network depends on the number of MCSs for the flows and the cardinality of the MCSs. Ideally, fewer MCSs with high cardinality are a sign of strong robustness. This is because fewer MCSs mean fewer possibilities for the flow to fail, and high MCS cardinality means several nodes must fail for that MCS to fail. The probability that all the elements of a large MCS fail simultaneously is minimal.

Though the inherent structural robustness of the topology is beyond the network operator's control in most cases, the operator can still improve network sovereignty by having maximum manufacturer diversity, even in the larger MCSs. For example, the average Flow-CSC score for Polska in Fig. 2 is nearly 1.0 for five manufacturers ($|M| = 5$). This is because the average MCS cardinality for Polska is 3.2, with ten MCSs in the network for different flows having a cardinality of 4. Hence, it is advisable to have more manufacturers in the network than the highest MCS cardinality in the network. This inference also agrees with the results in the other topologies.

3) Why do some flows have a Flow-CSC score equal to 0?
Some flows have Flow-CSC scores of 0. For example, CC, *Naga*, and the *Min cut set heuristic* for two manufacturers ($|M| = 2$) have Flow-CSC score of 0, shown by the outliers in Fig. 2. This occurs because some flows have at least one low cardinality MCS with nodes from the same manufacturer.However, this behavior is unavoidable when the number of manufacturers is low and the assignment is sub-optimal.

4) How should the network operator map the real manufacturers to the manufacturer IDs used in this study?
Fig. 3a shows the manufacturer(s) causing the most impactful failures. For example, for *CSC-ILP*, for the Polska topology, for four manufacturers, a combination of manufacturers M_1 and M_3 has the highest impact on failing, shown by the low Flow-CSC score of 0.5 for the red triangle in the middle grey box. Therefore, the operator must assign the most trustworthy manufacturers to the manufacturer IDs M_1 and M_3. Since there is no direct mapping between manufacturers in the physical world and the manufacturer-IDs used in this study, this is up to the operator to decide. Hence, this analysis helps the operator avoid the worst-case scenarios possible.

C. Guidelines to operators

Based on these results, we provide the following guidelines to operators.

- (G1) It is unsafe to purchase most components from one manufacturer. A fairly distributed manufacturer assignment is advisable.
- (G2) The number of manufacturers required depends on the average MCS cardinality. If the number of manufacturers exceeds the average MCS cardinality, the network is more sovereign.
- (G3) Having several manufacturers might be difficult in a real-life scenario due to a limited number of manufacturers in the market and interoperability issues.
- (G4) Manufacturer dependency can be avoided by using the CSC score and *CSC-ILP* for manufacturer assignment.

D. Network-CSC score and CSC-ILP Limitations

There are several networking devices from different manufacturers that use a particular subcomponent from the same subcomponent manufacturer. Hence, the Network-CSC score and *CSC-ILP*, currently focusing on component-level sovereignty, need to be further adapted to consider subcomponent-sharing. Moreover, the Network-CSC score does not consider the probability of a particular manufacturer failing. Modeling the trustworthiness of a manufacturer remains an open challenge. Additionally, the *CSC-ILP* must be extended to optimize for other dependability attributes discussed in [8].

VI. Conclusion and Outlook

This work introduced the novel CSC score metric to quantify and compare the network sovereignty of different manufacturer assignments and networks. We proposed the *CSC-ILP* to minimize the impact of manufacturer failures in networks by removing dependencies on the manufacturer(s). The results show the CSC score's validity as a sovereignty metric and *CSC-ILP*'s solution as a fast and scalable solution

to maximize network sovereignty. In the future, we aim to combine sovereignty and reliability for robust network designs.

REFERENCES

[1] M. Zhang, "Data center fires: A detailed breakdown with 22 examples," *Dgtl Infra*, 2023. [Online]. Available: https://dgtlinfra.com/data-center-fires/
[2] B. Fung, "We finally know what caused the global techoutage - and how much it cost," *CNN Business*, 2024. [Online]. Available: https://edition.cnn.com/2024/07/24/tech/crowdstrike-outage-cost-cause/index.html
[3] CVE Details, "CVE List." [Online]. Available: https://www.cvedetails.com/
[4] D. Tipper and P. Krishnamurthy, "Digital sovereignty and resilience," *Available at SSRN 4178621*, August 2022. [Online]. Available: https://ssrn.com/abstract=4178621
[5] S. Janardhanan *et al.*, "Network sovereignty: A novel metric and its application on network design," *IEEE Trans. Rel.*, pp. 1–15, 2024.
[6] British Broadcasting Corporation, "*US* telcos ordered to 'rip and replace' Huawei components," *BBC News*, Dec. 2020. [Online]. Available: https://www.bbc.com/news/business-55269879
[7] H. Li *et al.*, "Interpretation of network sovereignty," *Co-governed Sovereignty Netw.: Legal Basis and Its Prototype & Appl. with MIN Archit.*, pp. 29–60, 2021.
[8] A. Avizienis *et al.*, "Basic concepts and taxonomy of dependable and secure computing," *IEEE Trans. Depend. Sec. Comput.*, vol. 1, no. 1, pp. 11–33, 2004.
[9] J. Edler *et al.*, "Technology sovereignty: From demand to concept [Technologiesouveränität: Von der Forderung zum Konzept]," Fraunhofer Inst. for Syst. and Innov. Res. (ISI), Tech. Rep., 2020.
[10] S. Janardhanan *et al.*, "Improving network sovereignty - a minimal cut set approach," in *24th Int. Conf. on Transparent Opt. Netw. (ICTON) Demo Zone*, 2024.
[11] A. Newell *et al.*, "Increasing network resiliency by optimally assigning diverse variants to routing nodes," *IEEE Trans. Depend. Sec. Comput.*, vol. 12, no. 6, pp. 602–614, 2015.
[12] F. Ghawash and W. Abbas, "Leveraging diversity for achieving resilient consensus in sparse networks," *8th IFAC Workshop on Distrib. Estimation and Control in Netw. Syst. NECSYS*, vol. 52, no. 20, pp. 339–344, 2019.
[13] S. Orlowski *et al.*, "SNDlib 1.0—survivable network design library," *Netw.: An Int. J.*, vol. 55, no. 3, pp. 276–286, 2010.
[14] Gurobi Optimization, LLC, "Gurobi Optimizer Reference Manual." [Online]. Available: https://www.gurobi.com
[15] S. Neumayer and E. Modiano, "Network reliability with geographically correlated failures," in *Proc. of IEEE INFOCOM*, 2010, pp. 1–9.

VII. BIOGRAPHY SECTION

Shakthivelu Janardhanan joined the Chair of Communication Networks, Technical University of Munich (TUM) as a doctoral candidate in 2021 focusing on network dependability.

Ritanshi Agarwal joined the Chair of Communication Networks at Universität der Bundeswehr (UniBw) München as a research associate in 2024, focusing on resilient optical access networks.

Wolfgang Kellerer is a Full Professor at TUM, heading the Chair of Communication Networks. He's an editor for IEEE Transactions on Network and Service Management and Network Virtualization for IEEE Communications Surveys and Tutorials.

Carmen Mas-Machuca is a Full professor at UniBw, heading the Chair of Communication Networks. She's an editor for IEEE Transactions on Network and Service Management, OSA Journal on Optical Communications and Networking, and IEEE Communications Magazine.